

Class12

Rongyu Chen(A17708249)

2026-05-11

Section 1. Proportion of G|G in a population

Downloaded a CSV file from Ensembl < https://uswest.ensembl.org/Homo_sapiens/Variation/Sample?db=cor39975098;v=1s8067378;vdb=variation;vf=88214830#373531_tablePanel > Here we read this CSV file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

```
## Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
## 1 NA19648 (F) A|A ALL, AMR, MXL -
## 2 NA19649 (M) G|G ALL, AMR, MXL -
## 3 NA19651 (F) A|A ALL, AMR, MXL -
## 4 NA19652 (M) G|G ALL, AMR, MXL -
## 5 NA19654 (F) G|G ALL, AMR, MXL -
## 6 NA19655 (M) A|G ALL, AMR, MXL -
## Mother
## 1 -
## 2 -
## 3 -
## 4 -
## 5 -
## 6 -
```

```
table(mxl$Genotype..forward.strand.)
```

```
##
## A|A A|G G|A G|G
## 22 21 12 9
```

```
table(mxl$Genotype..forward.strand.)/nrow(mxl) * 100
```

```
##
## A|A A|G G|A G|G
## 34.3750 32.8125 18.7500 14.0625
```

Now lets look at a different population. I picked GBR

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

Find proportion of G|G

```
round(table(gbr$Genotype..forward.strand.) / nrow(gbr)*100, 2)
```

```
##
## A|A A|G G|A G|G
## 25.27 18.68 26.37 29.67
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the MKL population. Lets now dig into this further.

Section 4: Population Scale Analysis

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

How many samples do we have?

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

```
##      sample geno      exp
## 1 HG00367   A/G 28.96038
## 2 NA20768   A/G 20.24449
## 3 HG00361   A/A 31.32628
## 4 HG00135   A/A 34.11169
## 5 NA18870   G/G 18.25141
## 6 NA11993   A/A 32.89721
```

```
nrow(expr)
```

```
## [1] 462
```

```
table(expr$geno)
```

```
##
## A/A A/G G/G
## 108 233 121
```

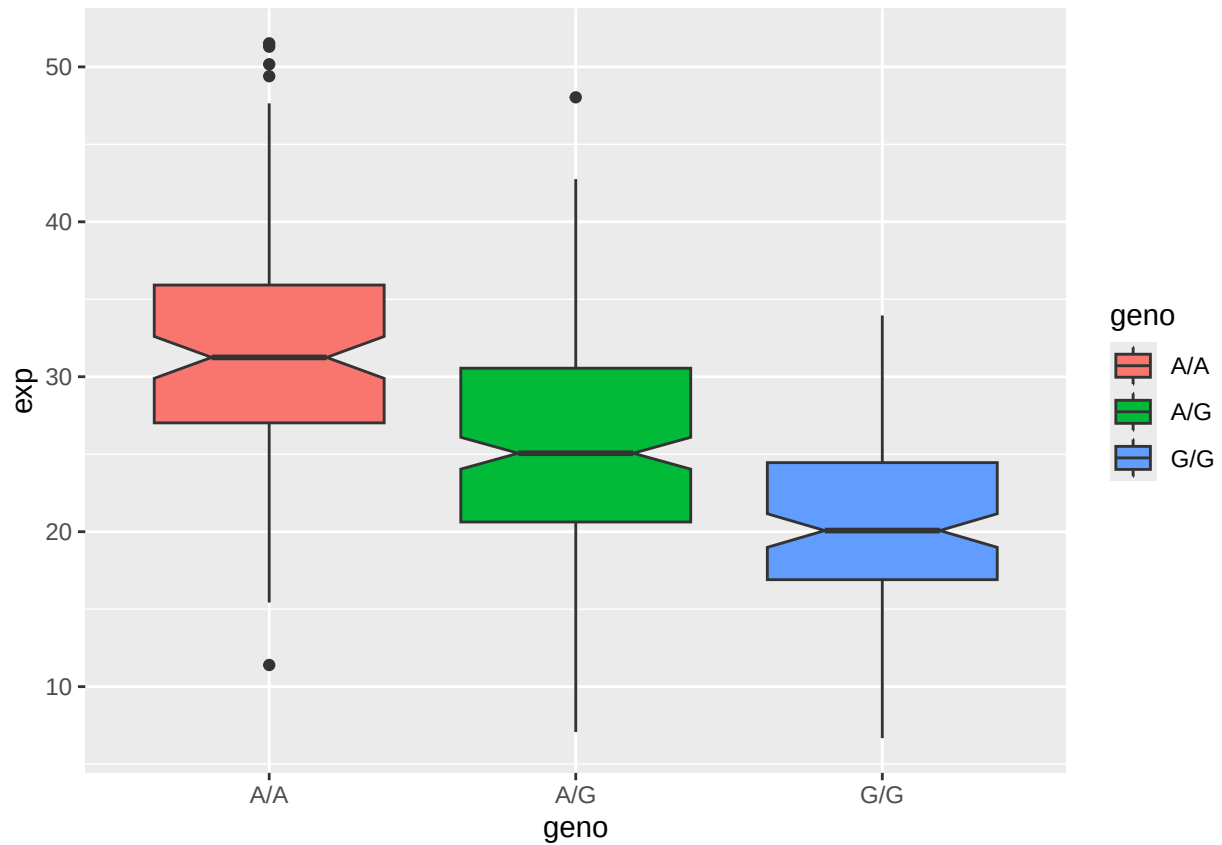
```
median_expression <- tapply(expr$exp, expr$geno, median)
print(median_expression)
```

```
##      A/A      A/G      G/G
## 31.24847 25.06486 20.07363
```

The sample sizes for each genotype are A/A: 108, A/G: 233, and G/G: 121. The calculated median expression levels show that A/A is 31.25, A/G is 25.06, and G/G is 20.07.

```
library(ggplot2)
```

```
ggplot(expr) + aes(x=geno, exp, fill=geno) +
geom_boxplot(notch=TRUE)
```



The A/A genotype exhibits a noticeably higher median expression level compared to the G/G genotype. This suggests that the SNP does affect the expression of ORMDL3, as the homozygous G/G genotype is associated with reduced expression values relative to A/A. So the SNP affects the expression of ORMDL3. Specifically, the G allele is associated with reduced expression of the gene compared to the A allele.